EE582 Homework 01 Report

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1 Question 1

A shunt harmonic filter (series RLC) is shown in Fig. 1(a). It is designed for a three-phase 13.8 kV, 60 Hz system to shunt the 7th harmonic current. The circuit parameters are:

- $r_L = 1\ \Omega$
- $L = 19$ mH
- $C = 8.2\ \mu$F

Step 1: Write the KVL equation:

$$V_{\text{dc}} - r_L i_L - L\frac{di_L}{dt} - v_C = 0$$

Or equivalently:

$$L\frac{di_L}{dt} + r_L i_L + v_C = V_{\text{dc}}$$

With $i_L = C\frac{dv_C}{dt}$ and $v_C = \frac{1}{C}\int i_L dt$.
Step 2: Standard second-order ODE:

$$\frac{d^2 i_L}{dt^2} + \frac{r_L}{L}\frac{di_L}{dt} + \frac{1}{LC}i_L = 0$$

Step 3: Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Step 4: Calculate parameters:

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{19\times10^{-3}\times 8.2\times10^{-6}}} = 2533.5 \text{ rad/s}$$
$$\zeta = \frac{r_L}{2\sqrt{\frac{L}{C}}} = \frac{1}{2\sqrt{\frac{19\times10^{-3}}{8.2\times10^{-6}}}} = 0.0104$$
$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2533.3 \text{ rad/s}$$

Step 5: Quality factor:

$$Q_{\text{factor}} = \frac{\omega_n L}{r_L} = \frac{2533.5\times 0.019}{1} = 48.136$$

Step 6: Eigenvalues:

$$\lambda = -\zeta\omega_n \pm j\omega_d = -26.316 \pm 2533.3j$$

Step 7: Reactive power:

$$Q_{\text{kVAr}} = -602.04 \text{ kVAr}$$

All calculations confirm the system is very underdamped ($\zeta = 0.0104$).

Summary of Key Results:

- Natural frequency: $\boxed{\omega_n = 2533.5 \text{ rad/s}}$
- Damping ratio: $\boxed{\zeta = 0.0104}$
- Damped frequency: $\boxed{\omega_d = 2533.3 \text{ rad/s}}$
- Quality factor: $\boxed{Q = 48.136}$
- Eigenvalues: $\boxed{-26.316 \pm 2533.3j}$
- Reactive power: $\boxed{-602.04 \text{ kVAr}}$

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